

WHAT σ -EQUIVARIANCE GIVES FOR FREE, AND WHY THE RANGE PROJECTIONS CANNOT GIVE THE REST

RUQING CHEN

ABSTRACT. We examine the proposal to prove Cartan uniqueness for $(\mathcal{A}_{K,S}, \sigma)$ by spectral means alone, and separate what it yields from what it cannot.

On the positive side, σ -equivariance already gives more than Paper XIII of the series (its Theorem 2.5) extracted from it, and gives it without any Cartan input. The eigenvalue group $\mathrm{Sp}_p(\sigma)$ of the flow is the subgroup of $\mathbb{R}_{>0}$ generated by $\{\mathrm{Np}\}_{p \in S}$; and the entire KMS bundle is defined from $(\mathcal{A}, \sigma)$ alone, so a σ -equivariant isomorphism transports it. Consequently the function $\beta \mapsto G_S/\Xi_\beta^\perp$, up to homeomorphism, is an invariant: in particular $|\Xi_\beta|$ wherever it is finite, the transition locus, and β_c . *No Cartan subalgebra is needed for any of this.* What Cartan uniqueness is needed for is exactly one thing: the norm *multiset*, since $\mathrm{Sp}_p(\sigma)$ sees only the group generated by the norms and is blind to multiplicity — two equal-norm primes and one prime contribute the same subgroup.

On the negative side, the proposed route cannot deliver it, and the obstruction is at Step 3 rather than Step 2. The semilattice $\mathcal{E} = \{e_a\}$ of range projections does *not* generate $C(Y_S)$: since $e_a = \mathbf{1}_{aY_S}$ is the condition $v_p(x) \geq a_p$, these functions depend only on the valuations and take equal values at $[x, g]$ and $[x, g']$ for all $g, g' \in G_S$. They do not separate points, and $C^*(\mathcal{E}) \cong C(\overline{\mathbb{N}}^S)$ is a proper subalgebra, blind to the whole G_S -direction — which by the exact sequence $1 \rightarrow U_S/\overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \mathrm{Cl}_S^+ \rightarrow 1$ is most of Y_S . A complete spectral recovery of the range projections would therefore still not give $C(Y_S)$, and this failure is independent of whether Step 2 holds.

On Step 2 itself the verdict is negative, and no ergodic theory is needed. For equal-norm primes $\mathfrak{p}_1, \mathfrak{p}_2$ the element $v = \mu_{\mathfrak{p}_1} e_{\mathfrak{p}_1} + \mu_{\mathfrak{p}_2} (1 - e_{\mathfrak{p}_1})$, with $e_{\mathfrak{p}_1} = \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^*$, is an isometry of pure frequency — $\mu_{\mathfrak{p}_1}^* (1 - e_{\mathfrak{p}_1}) = 0$, so the cross terms vanish identically — and it is not diagonal. Mixed isometries therefore exist for every equal-norm pair. The general criterion, for $v = \mu_{\mathfrak{p}_1} p + \mu_{\mathfrak{p}_2} (1 - p)$ with $g = \mathfrak{p}_2 \mathfrak{p}_1^{-1}$ and $U = \mathrm{supp}(1 - p)$, is $v^* v = 1 \iff g(U \cap \mathfrak{p}_1 Y_S) \subseteq U$; whether the measure-preserving element g ($\mathrm{Ng} = 1$) is ergodic on $(\tilde{Y}_S, \nu_\beta)$ remains open, but it no longer governs the existence question.

The gauge-equivariant theorem of Paper XIV (its Theorem 4.2) therefore remains the strongest unconditional statement, and the norm multiset remains the single quantity that σ -equivariance alone has not been shown to recover.

1. SPECTRAL SUBSPACES AND WHAT THEY SEE

We use the notation of [4]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, spanned by the monomials $\mu_a f \mu_b^*$, with $\sigma_t(\mu_a) = \mathrm{Na}^{it} \mu_a$; $\tilde{Y}_S$ and the quasi-invariant measures ν_β are as there. Papers XIII and XIV of the series, referred to by numeral, are unpublished companions; every fact used from them is restated in place. For $\lambda > 0$ let

$$\mathcal{A}(\lambda) = \{x \in \mathcal{A}_{K,S} : \sigma_t(x) = \lambda^{it} x \ \forall t\} = \overline{\mathrm{span}}\{\mu_a f \mu_b^* : \mathrm{Na}/\mathrm{Nb} = \lambda, f \in C(Y_S)\}$$

be the spectral subspaces of the flow in the sense of Arveson [1].

Proposition 1.1 (The point spectrum). $\mathrm{Sp}_p(\sigma) = \{\lambda > 0 : \mathcal{A}(\lambda) \neq 0\}$ is the subgroup of $(\mathbb{R}_{>0}, \times)$ generated by $\{\mathrm{Np}\}_{p \in S}$. It is an invariant of the C^* -dynamical system, requiring no Cartan subalgebra. The Arveson spectrum proper — the closed support of the spectral subspaces

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XV-spectral/](#).

— is the closure of this subgroup, hence all of $\mathbb{R}_{>0}$ as soon as two of the $\log \mathbf{Np}$ have irrational ratio; the eigenvalue group $\mathrm{Sp}_p(\sigma)$, not its closure, is the invariant that retains the information.

Proof. Every $\lambda = \mathbf{Na}/\mathbf{Nb}$ is realized by the eigenvector $\mu_a f \mu_b^*$, and these ratios form exactly the subgroup generated by the $\mathbf{Np}$. Conversely let $0 \neq x \in \mathcal{A}(\lambda)$ and pick a functional φ with $\varphi(x) \neq 0$. Then $t \mapsto \varphi(\sigma_t(x)) = \lambda^{it} \varphi(x)$ is the uniform limit of the functions $t \mapsto \varphi(\sigma_t(x_n))$ for finite sums x_n of monomials, which are trigonometric polynomials with frequencies in $\log \langle \mathbf{Np} \rangle$; the Bohr–Fourier coefficient at $\log \lambda$ is continuous for uniform convergence, vanishes on every x_n if $\lambda \notin \langle \mathbf{Np} \rangle$, and equals $\varphi(x) \neq 0$ on the limit. So $\lambda \in \langle \mathbf{Np} \rangle$. Invariance is immediate from $\Phi \circ \sigma_t^1 = \sigma_t^2 \circ \Phi$. $\square$

Remark 1.2 (What the spectrum misses). $\mathrm{Sp}_p(\sigma)$ records the group generated by the norms, not the multiset. If $\mathbf{p}_1 \neq \mathbf{p}_2$ have $\mathbf{Np}_1 = \mathbf{Np}_2 = q$, the subgroup contributed is $q^{\mathbb{Z}}$, the same as for a single prime of norm q . Multiplicity is exactly the information that the spectrum discards, and by §2 it is the only piece of the reconstruction that is not already free.

2. WHAT σ -EQUIVARIANCE GIVES FOR FREE

Proposition 2.1 (The KMS bundle is intrinsic). *The set $\mathrm{KMS}_\beta(\mathcal{A}, \sigma)$ is defined from $(\mathcal{A}, \sigma)$ alone. Hence a σ -equivariant isomorphism Φ induces, for every β , an affine homeomorphism $\mathrm{KMS}_\beta(\mathcal{A}_1, \sigma_1) \rightarrow \mathrm{KMS}_\beta(\mathcal{A}_2, \sigma_2)$, with no hypothesis on Cartan subalgebras.*

Theorem 2.2 (Free consequences). *Let $\Phi : (\mathcal{A}_{K_1, S_1}, \sigma_1) \rightarrow (\mathcal{A}_{K_2, S_2}, \sigma_2)$ be a σ -equivariant isomorphism. Then, unconditionally — for all number fields, all prime sets, separated or not:*

- (1) $\mathrm{Sp}_p(\sigma_1) = \mathrm{Sp}_p(\sigma_2)$, so the subgroups of $\mathbb{R}_{>0}$ generated by the norms agree;
- (2) for every $\beta > 0$ the spaces $G_{S_1}/\Xi_\beta(K_1, S_1)^\perp$ and $G_{S_2}/\Xi_\beta(K_2, S_2)^\perp$ are homeomorphic, being the extreme boundaries of the corresponding simplices $\mathrm{Prob}(G_S/\Xi_\beta^\perp)$ of [4, Thm. 3.6]; in particular, taking any $\beta > \beta_c$, the compact spaces G_{S_1} and G_{S_2} are homeomorphic;
- (3) in particular $|\Xi_\beta(K_1, S_1)| = |\Xi_\beta(K_2, S_2)|$ whenever these are finite;
- (4) the transition loci coincide, and $\beta_c(S_1) = \beta_c(S_2)$.

Proof. (1) is Proposition 1.1. (2)–(4) follow from Proposition 2.1: the extreme boundary of a Bauer simplex is a homeomorphism invariant, and by [4, Thm. 3.6] it is $G_S/\Xi_\beta^\perp$, which for $\beta > \beta_c$ is G_S itself since $\zeta_{K,S}(\beta) < \infty$ forces $\Xi_\beta = \widehat{G_S}$; the transition locus is the set of β at which the simplex changes, and β_c is determined with it — this is the argument of Theorem 2.5(5) of Paper XIII of the series, which uses only the transport of KMS states available here and no Cartan input. $\square$

Remark 2.3 (The gap is exactly the multiset). Comparing Theorem 2.2 with Theorem 2.5 of Paper XIII, the conclusions there not obtained here are the norm-preserving bijection $S_1 \rightarrow S_2$ of its parts (2)–(3) and, in its conditional part (6), the matching of Frobenius classes and the transport of $\{\Xi_\beta\}$ as subgroups of $\widehat{G_S}$ — here only the cardinalities and the quotient spaces come for free. The narrow class number is a conclusion in neither place: by Remark 3.2 of Paper XIII, h_S^+ is recovered by no argument there, and nothing here adds one. Everything else — G_S as a compact space, the chain $\{\Xi_\beta\}$ in cardinality, the transition locus, β_c — comes free from σ -equivariance. So the Cartan question is not a prerequisite for the arithmetic reconstruction in general; it is a prerequisite for one specific piece of it, the multiplicity of the norms, and by Remark 1.2 that is precisely what $\mathrm{Sp}_p(\sigma)$ cannot see.

3. MIXED ISOMETRIES EXIST: STEP 2 FAILS

Assume $\mathbf{p}_1 \neq \mathbf{p}_2$ in S with $\mathbf{Np}_1 = \mathbf{Np}_2 = q$, and put $g = \mathbf{p}_2 \mathbf{p}_1^{-1} \in I_S$.

Lemma 3.1 (Nica relation). *For coprime $\mathfrak{p}_1, \mathfrak{p}_2$ one has $\mu_{\mathfrak{p}_1}^* \mu_{\mathfrak{p}_2} = \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$.*

Proof. $\mathfrak{p}_1 \vee \mathfrak{p}_2 = \mathfrak{p}_1 \mathfrak{p}_2$, and the Nica covariance relation [2] $\mu_a^* \mu_b = \mu_{a^{-1}(a \vee b)} \mu_{b^{-1}(a \vee b)}^*$ gives the claim. $\square$

Proposition 3.2 (Mixed isometries exist; the general criterion). *Let $p \in C(Y_S)$ be a projection, $U = \text{supp}(1 - p)$, and $v = \mu_{\mathfrak{p}_1} p + \mu_{\mathfrak{p}_2} (1 - p)$. Then*

$$v^* v = 1 + \mathbf{1}_{U^c \cap g(U \cap \mathfrak{p}_1 Y_S)} \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^* + \text{adjoint}, \quad \text{so} \quad v^* v = 1 \iff g(U \cap \mathfrak{p}_1 Y_S) \subseteq U,$$

where g acts as the partial homeomorphism $\mathfrak{p}_1 Y_S \rightarrow \mathfrak{p}_2 Y_S$. In particular the condition is vacuous when $U \cap \mathfrak{p}_1 Y_S = \emptyset$: for $p = e_{\mathfrak{p}_1}$, i.e. $U = Y_S \setminus \mathfrak{p}_1 Y_S$, where $e_{\mathfrak{p}_1} = \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^* = \mathbf{1}_{\mathfrak{p}_1 Y_S}$,

$$v = \mu_{\mathfrak{p}_1} e_{\mathfrak{p}_1} + \mu_{\mathfrak{p}_2} (1 - e_{\mathfrak{p}_1})$$

is an isometry, since $\mu_{\mathfrak{p}_1}^* (1 - e_{\mathfrak{p}_1}) = \mu_{\mathfrak{p}_1}^* - \mu_{\mathfrak{p}_1}^* \mu_{\mathfrak{p}_1} \mu_{\mathfrak{p}_1}^* = 0$ kills the cross terms outright, and it is homogeneous for σ of frequency q but not of a single gauge degree, hence not of the form $\mu_{\mathfrak{p}} f$. Mixed isometries of pure frequency exist for every equal-norm pair, with no further hypothesis.

Proof. Expanding and using Lemma 3.1, $v^* v = p + (1 - p) + p \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^* (1 - p) + \text{adjoint}$. Pushing the functions through the partial isometry $\mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$, which implements $g : \mathfrak{p}_1 Y_S \rightarrow \mathfrak{p}_2 Y_S$, turns the third term into $\mathbf{1}_{U^c} \mathbf{1}_{g(U \cap \mathfrak{p}_1 Y_S)} \mu_{\mathfrak{p}_2} \mu_{\mathfrak{p}_1}^*$, which lies in the spectral subspace of gauge degree $\mathfrak{p}_2 - \mathfrak{p}_1 \neq 0$ and so is linearly independent of $C(Y_S)$; it vanishes iff $U^c \cap g(U \cap \mathfrak{p}_1 Y_S) = \emptyset$, i.e. $g(U \cap \mathfrak{p}_1 Y_S) \subseteq U$. The special case is the display, and v has gauge degrees $\delta_{\mathfrak{p}_1}$ and $\delta_{\mathfrak{p}_2}$ both occurring, while any $\mu_{\mathfrak{p}} f$ is homogeneous. $\square$

Remark 3.3 (Status). Step 2 of the proposed route — “every isometry of pure frequency is diagonal” — is therefore *false*, by a two-line example, and the failure needs no ergodic theory: the projection $e_{\mathfrak{p}_1}$, being a range projection, is triggered by the valuations alone, and mixing over it costs nothing. An earlier draft of this note claimed that mixed isometries exist *iff* the element $g = \mathfrak{p}_2 \mathfrak{p}_1^{-1}$ fails to be ergodic on $(\tilde{Y}_S, \nu_\beta)$; both implications fail — the example above uses no invariant set, and non-ergodicity produces measurable, not clopen, invariant sets. What survives of the ergodic picture is this: $Ng = 1$, so g preserves each ν_β (the Radon–Nikodym cocycle of the I_S -action being $(Ng)^{-\beta}$), the full group I_S acts ergodically by the Kolmogorov zero–one law [4, Lem. 3.1], and whether a *single* such g — shifting $v_{\mathfrak{p}_2}$ up, $v_{\mathfrak{p}_1}$ down, and translating G_S by $\sigma_{\mathfrak{p}_2} \sigma_{\mathfrak{p}_1}^{-1}$ — is ergodic remains open and of independent interest; but it no longer decides the existence of mixed isometries, which is settled affirmatively.

4. THE DECISIVE GAP: RANGE PROJECTIONS ARE BLIND TO G_S

Theorem 4.1 ($\mathcal{E}$ does not generate the diagonal). *Let $\mathcal{E} = \{e_{\mathfrak{a}} = \mathbf{1}_{\mathfrak{a} Y_S} : \mathfrak{a} \in J_S\}$. Then $\mathcal{E}$ does not separate the points of Y_S : for every $\mathfrak{a}$ and all $g, g' \in G_S$,*

$$e_{\mathfrak{a}}([x, g]) = e_{\mathfrak{a}}([x, g']).$$

Consequently $C^*(\mathcal{E}) \cong C(\overline{\mathbb{N}}^S)$, the algebra of continuous functions of the valuations alone, and is a proper subalgebra of $C(Y_S)$ whenever G_S is nontrivial.

Proof. $\mathfrak{a} Y_S = \{[x, g] : v_{\mathfrak{p}}(x) \geq a_{\mathfrak{p}} \forall \mathfrak{p}\}$, a condition on x only, so $e_{\mathfrak{a}}$ factors through the projection $Y_S \rightarrow \overline{\mathbb{N}}^S$, $[x, g] \mapsto (v_{\mathfrak{p}}(x))_{\mathfrak{p}}$. The $e_{\mathfrak{a}}$ generate the clopen algebra of $\overline{\mathbb{N}}^S$, whence the identification by Stone–Weierstrass; properness follows since $G_S \neq \{1\}$ for every $S \neq \emptyset$, by [4, (2.1)]. $\square$

Corollary 4.2 (Step 3 fails independently of Step 2). *Even if every isometry of pure frequency q were diagonal — and by Proposition 3.2 it is not — so that the range projections $e_{\mathfrak{p}}$ were spectrally recoverable, the algebra they generate is $C(\overline{\mathbb{N}}^S)$ and not $C(Y_S)$. The proposed route therefore cannot establish $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, the identity needed to bring Renault’s reconstruction [3] to bear, and this failure is located at Step 3, independently of Step 2.*

Remark 4.3 (How much is missed). By [4, (2.1)], G_S sits in $1 \rightarrow U_S/\overline{\mathcal{O}_{K,+}^\times} \rightarrow G_S \rightarrow \mathrm{Cl}_S^+ \rightarrow 1$ with $U_S = \prod_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^\times$ an infinite product of infinite profinite groups. The invisible direction is therefore not a small correction but a factor comparable in size to Y_S itself. Any route to $C(Y_S)$ must produce elements that vary along G_S , and the range projections, being defined by valuation inequalities, never do.

Remark 4.4 (What would have to replace it). Elements of $\mathcal{A}$ that vary along G_S do exist in abundance — all of $C(Y_S)$ does — but they are not distinguished by the flow, σ acting trivially on $C(Y_S)$. To characterize them one would need a construction using the multiplication of $\mathcal{A}$ against the spectral subspaces, for instance identifying $C(Y_S)$ inside $\mathcal{A}^\sigma$ as a distinguished maximal abelian subalgebra. That is the question left open in Remark 3.3 of Paper XIV, and the present note does not advance it; what it does is show that the range projections are not a route to it.

5. ASSESSMENT

Statement	Status	Reference
$\mathrm{Sp}_p(\sigma) = \langle \mathrm{Np} \rangle \leq \mathbb{R}_{>0}$	true, intrinsic	Prop. 1.1
$\mathrm{Sp}_p(\sigma)$ sees the norm multiset	false: no multiplicity	Rem. 1.2
KMS bundle intrinsic to $(\mathcal{A}, \sigma)$	true	Prop. 2.1
G_S , $\{ \Xi_\beta \}$, transition loci, β_c recovered	true, unconditionally	Thm. 2.2
h_S^+ recovered	not claimed (as in Paper XIII)	Rem. 2.3
norm multiset recovered	open; the only gap	Rem. 2.3
mixed isometries exist	true, unconditionally	Prop. 3.2
every pure-frequency isometry is diagonal	false	Prop. 3.2
mixed isometries $\iff g$ non-ergodic	false, both ways	Rem. 3.3
$\mathcal{E}$ generates $C(Y_S)$	false: blind to G_S	Thm. 4.1
spectral route to Cartan uniqueness	closed	Cor. 4.2
σ -equivariance $\Rightarrow$ gauge-equivariance	not established	—

Two conclusions, of opposite sign, and the positive one is the more useful.

The reconstruction of Paper XIII (its Theorem 2.5) was presented as resting on Cartan uniqueness throughout. Theorem 2.2 shows that most of it does not: the KMS bundle is manifestly intrinsic to $(\mathcal{A}, \sigma)$, and with it come G_S as a compact space, the cardinalities of the obstruction groups, the transition locus and the critical exponent, for every number field and every prime set, separated or not. The Cartan question is needed for one thing only, the multiplicity of the norms, which is exactly what the point spectrum discards.

The proposed spectral route to that last piece is closed twice over. Step 2 is false outright: mixed isometries of pure frequency exist for every equal-norm pair (Proposition 3.2). And even its truth would have delivered only the semilattice $\mathcal{E}$, which by Theorem 4.1 is blind to G_S , generating a proper subalgebra of the diagonal. The unconditional statement therefore remains Theorem 4.2 of Paper XIV, with gauge-equivariance as hypothesis.

Two questions. Is $g = \mathfrak{p}_2 \mathfrak{p}_1^{-1}$ ergodic on $(\tilde{Y}_S, \nu_\beta)$? By Remark 3.3 this no longer decides the existence of mixed isometries, but it is a clean self-contained problem about a single measure-preserving transformation of a product of geometric measures twisted by a group translation, and we have not decided it. And is there any construction recovering the G_S -direction of $C(Y_S)$ from $(\mathcal{A}, \sigma)$? Remark 4.4 explains why the obvious candidates do not, the flow being trivial on precisely the algebra one is trying to find.

REFERENCES

- [1] W. Arveson, *On groups of automorphisms of operator algebras*, J. Funct. Anal. 15 (1974), 217–243.

- [2] A. Nica, *C^* -algebras generated by isometries and Wiener–Hopf operators*, J. Operator Theory 27 (1992), 17–52.
- [3] J. Renault, *Cartan subalgebras in C^* -algebras*, Irish Math. Soc. Bull. 61 (2008), 29–63.
- [4] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com